Author result prop-norm (printed 2.5) of ahl-163-stochastic-nonlinear-wave-dynamics-on-compact-2023

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Editorial note. Local material id L42305, claim key doi:10.5802/ahl.163:prop-norm, lane-x3 S062 batch-2 assignment (CLAIMS six-line segment).

Editorial scope. Single face of a Centre Mersenne journal unit: the counted text is the author's own statement and proof copied verbatim from the pinned source; the card adds no macro of its own beyond declaring the author environment names.

Allowed standard prerequisites (editorial).

- The statement and its own proof as printed in the article. source lines 723-766. the counted unit and the argument carried out in the span

Proposition 0.1. *Let $\kappa : U \subset \mathbb{R}^d \to V \subset \mathcal{M}$ be a coordinate patch and $\chi \in C_0^\infty(V)$. For any $s \in \mathbb{R}$ and $1 \le p, q \le \infty$, there exist $c, C > 0$ such that for any $u \in C^\infty(\mathcal{M})$,*

$$c\|\chi u\|_{B^s_{p,q}(\mathcal{M})} \le \|\kappa^\star(\chi u)\|_{B^s_{p,q}(\mathbb{R}^d)} \le C\|u\|_{B^s_{p,q}(\mathcal{M})}. \tag{1}$$

Proof. First, observe that it is enough to establish the right-hand side inequality, since by duality it holds

$$\begin{aligned}\|\chi u\|_{B^s_{p,q}(\mathcal{M})} &= \sup_{\|\widetilde{u}\|_{B^{-s}_{p',q'}(\mathcal{M})} \le 1} \int_{\mathcal{M}} \chi u \cdot \widetilde{u} = \sup_{\|\widetilde{u}\|_{B^{-s}_{p',q'}(\mathcal{M})} \le 1} \int_{\mathbb{R}^d} \kappa^\star(\chi u) \cdot \kappa^\star(\widetilde{\chi}\widetilde{u}) \\ &\lesssim \sup_{\|\widetilde{u}\|_{B^{-s}_{p',q'}(\mathcal{M})} \le 1} \|\kappa^\star(\chi u)\|_{B^s_{p,q}(\mathbb{R}^d)} \|\kappa^\star(\widetilde{\chi}\widetilde{u})\|_{B^{-s}_{p',q'}(\mathbb{R}^d)} \lesssim \|\kappa^\star(\chi u)\|_{B^s_{p,q}(\mathbb{R}^d)},\end{aligned}$$

where in the last step we used the right-hand side inequality in (1). This shows that the left-hand side inequality follows from the right-hand side one.

We thus need to estimate

$$\sum_{N \in 2^{\mathbb{Z}_+}} N^{sq} \|\theta_N(D)\kappa^\star(\chi u)\|^q_{L^p(\mathbb{R}^d)},$$

where $\{\theta_N\}_{N\in 2^{\mathbb{Z}_+}}$ is an inhomogeneous dyadic partition of unity in $\mathbb{R}^d$. We first take a fattened version $\widetilde{\psi}_{N_1^2}$ of $\psi_{N_1^2}$, where $\psi_{N_1^2}$ is the multiplier in the definition of $\mathbf{P}_{N_1}$, and decompose

$$\begin{aligned}\theta_N(D)\kappa^\star(\chi u) &= \sum_{N_1\in 2^{\mathbb{Z}_+}} \theta_N(D)\kappa^\star\left(\chi u_{N_1}\right)\\ &= \sum_{N_1\sim N} \theta_N(D)\kappa^\star(\chi u_{N_1}) + \sum_{N_1\not\sim N} \theta_N(D)\kappa^\star\left(\chi\widetilde{\psi}_{N_1^2}(-\Delta_{\mathfrak{g}})u_{N_1}\right),\end{aligned}$$

where $u_{N_1}=\psi_{N_1^2}(-\Delta_{\mathfrak{g}})u$. To bound the terms in the second sum above, we have the following lemma.

Lemma 0.1. *Let κ and χ as in Proposition 0.1. Then for any $u\in L^p(\mathcal{M})$, $p\geq 1$, and any $N,N_1\in 2^{\mathbb{Z}_+}$ with $(N\vee N_1)\gg(N\wedge N_1)$, we have for arbitrary $B>0$:*

$$\left\|\theta_N(D)\kappa^\star\left(\chi\widetilde{\psi}_{N_1^2}(-\Delta_{\mathfrak{g}})u\right)\right\|_{L^p(\mathbb{R}^d)} \lesssim (N\vee N_1)^{-B}\|u\|_{L^p(\mathcal{M})}. \tag{2}$$

With this lemma at hand, we can finish establishing the right-hand side inequality in (1). Indeed, for the almost diagonal terms, we have from Minkowski's inequality, the uniform boundedness of the Littlewood–Paley projectors $\theta_N(D)$ on $L^p(\mathbb{R}^d)$, and Hölder's inequality with Fubini's theorem that

$$\begin{aligned}\|\mathbf{1}_{N_1\sim N}N^s\theta_N(D)\kappa^\star(\chi u_{N_1})\|_{\ell^q_N(2^{\mathbb{Z}_+})L^p(\mathbb{R}^d)\ell^1_{N_1}(2^{\mathbb{Z}_+})}&\\ \lesssim \|\mathbf{1}_{N_1\sim N}N^s u_{N_1}\|_{\ell^q_{N_1}(2^{\mathbb{Z}_+})\ell^q_N(2^{\mathbb{Z}_+})L^p(\mathcal{M})}&\\ \lesssim \|u\|_{B^s_{p,q}(\mathcal{M})},&\end{aligned}$$

while for the off-diagonal terms we have from Minkowski's inequality and Lemma 0.1 applied to u_{N_1} with $B>2|s|$:

$$\begin{aligned}\left\|\mathbf{1}_{N_1\not\sim N}\theta_N(D)\kappa^\star\left(\chi\widetilde{\psi}_{N_1^2}(-\Delta_{\mathfrak{g}})u_{N_1}\right)\right\|_{\ell^q_N(2^{\mathbb{Z}_+})L^p(\mathbb{R}^d)\ell^1_{N_1}(2^{\mathbb{Z}_+})}&\\ \lesssim \left\|\mathbf{1}_{N_1\not\sim N}N^s(N\vee N_1)^{-B}u_{N_1}\right\|_{\ell^q_N(2^{\mathbb{Z}_+})L^p(\mathcal{M})\ell^1_{N_1}(2^{\mathbb{Z}_+})}&\\ \lesssim \left\|N_1^{|s|-B}u_{N_1}\right\|_{\ell^1_{N_1}(2^{\mathbb{Z}_+})L^p(\mathcal{M})} \lesssim \|u\|_{B^s_{p,q}(\mathcal{M})}.&\end{aligned}$$

This concludes the proof of Proposition 0.1, assuming Lemma 0.1. □